

CONTINUOUS ASSESSMENT 6

Multivariate Analysis

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Abstract
An exploration of MANOVA

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Question 1

We now have a dataset with 4 time periods, and we want to check if there are in the parameters of the multivariate normal distribution for the 4 time periods are the same. We will use Box's M test to check for homogeneity of covariance matrices. We will test the null hypothesis:

$$H_0 : \Sigma_1 = \Sigma_2 = \Sigma_3 = \Sigma_4$$

Against the alternative hypothesis:

$$H_a : \text{At least one of the covariance matrices is different}$$

Undertaking the Box's M test, we find that the p-value is 0.2483, which suggests that we fail to reject the null hypothesis of the homogeneity of covariance matrices. We are able to proceed to the next step of performing a one way MANOVA to check if the mean vectors of the 4 time periods are the same. We test the null hypothesis:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

Against the alternative hypothesis:

$$H_a : \text{At least one of the mean vectors is different}$$

This yields the following MANOVA table:

	Df	Wilks' Lambda	Approx F	Num Df	Den Df	Pr(>F)
groups	4	0.6635858	3.900928	16	434.4548	7.01e-07
Residuals	145	NA	NA	NA	NA	NA

Undertaking the one way MANOVA, we find that the p-value is 7.01×10^{-7} , which suggests that we reject the null hypothesis of the mean vectors being the same across the 4 time periods. There is exceptionally strong evidence to suggest that at least one of the mean vectors is different across the 4 time periods. We look at the individual ANOVA tables for each of the 4 variables to determine which of the variables are different across the 4 time periods.

Table 2: Summary of ANOVA for MaxBreadth

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
groups	4	502.8267	125.7067	5.954613	0.0001826317
Residuals	145	3061.0667	21.1108	NA	NA

Table 3: Summary of ANOVA for BasHeight

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
groups	4	229.9067	57.47667	2.44742	0.04896988
Residuals	145	3405.2667	23.48460	NA	NA

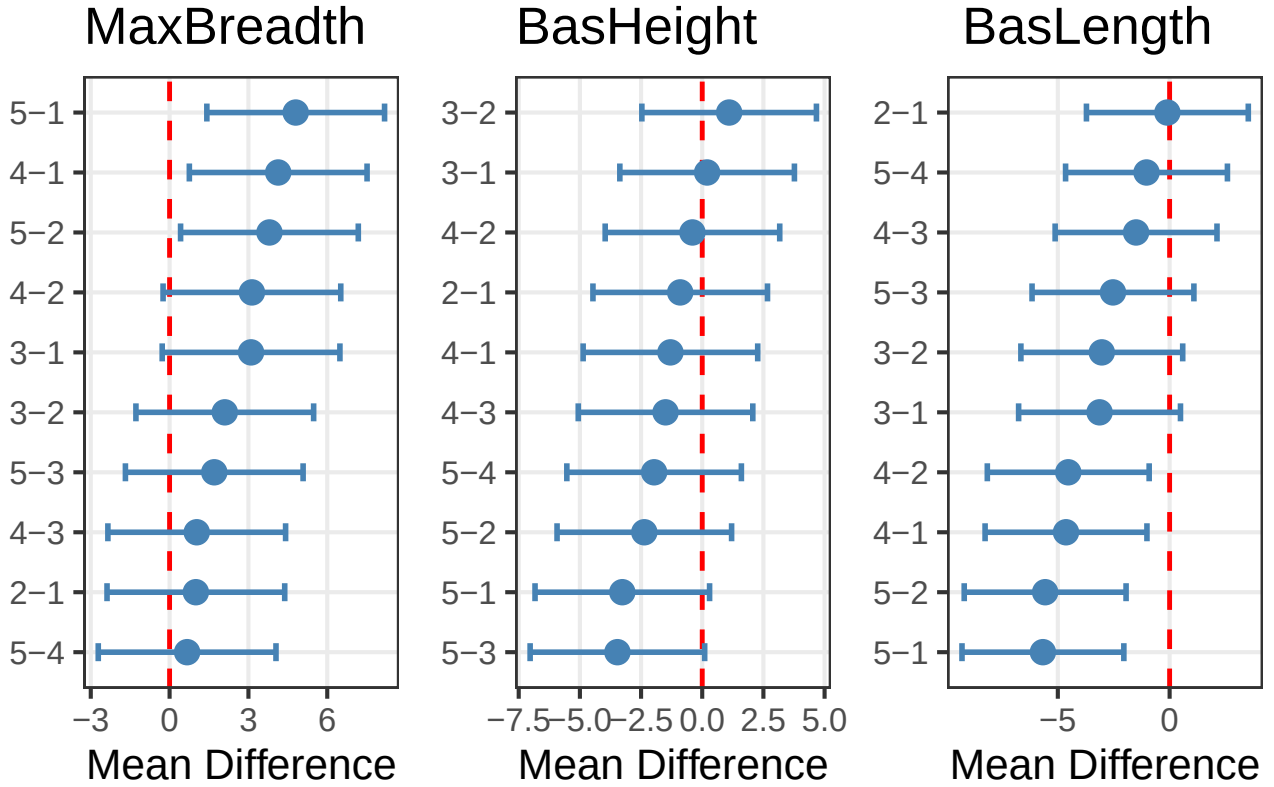
Table 4: Summary of ANOVA for BasLength

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
groups	4	803.2933	200.82333	8.305665	4.6364e-06
Residuals	145	3505.9667	24.17908	NA	NA

Table 5: Summary of ANOVA for NasHeight

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
groups	4	61.200	15.30000	1.506997	0.2031788
Residuals	145	1472.133	10.15264	NA	NA

This suggest that there is differences in the means for variable MaxBreadth, BasHeight, and BasLength across the 4 time periods. We now check if the pairwise differences in the means for these variables are significant across the 4 time periods using a post-hoc analysis with Bonferroni correction.



We have that the differences are significant for the time periods where the confidence intervals contain 0.

Question 2

a)

Table 6: MANOVA Table for Fabric Wear

	Df	Pillai's Trace	Approx F	Num Df	Den Df	Pr(>F)
Proportion	2	0.9981	1.6602	6	10	0.2282
Filler	1	0.9945	240.2160	3	4	0.0001
Proportion:Filler	2	0.9265	1.4385	6	10	0.2911
Residuals	6	NA	NA	NA	NA	NA

A two-factor MANOVA was conducted to assess the effects of filler type and filler proportion on fabric wear measured over three successive periods. Using Pillai's trace, the above multivariate results indicate that filler type has a statistically significant overall effect on the joint response vector of wear measurements ($p < 0.001$), whereas filler proportion ($p = 0.228$) and the interaction between filler type and proportion ($p = 0.291$) are not statistically significant when the three periods are considered jointly. Thus, in an overall multivariate sense, the main driver of differences in fabric wear is the type of filler used.

The individual ANOVA results, however, show that the influence of the factors varies across the three time periods;

Table 7: ANOVA Table for Fabric Wear - First Period

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Proportion	2	3936.5	1968.25	6.5066	0.0314
Filler	1	192.0	192.00	0.6347	0.4560
Proportion:Filler	2	1020.5	510.25	1.6868	0.2623
Residuals	6	1815.0	302.50	NA	NA

For the first 1000 revolutions (y_1), there is a significant effect of filler proportion ($p = 0.031$), while filler type and the interaction are not significant. This suggests that early fabric wear is primarily associated with the proportion of filler, with wear tending to increase as the proportion rises.

Table 8: ANOVA Table for Fabric Wear - Second Period

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Proportion	2	3418.167	1709.0833	13.2316	0.0063
Filler	1	33496.333	33496.3333	259.3265	0.0000
Proportion:Filler	2	2754.167	1377.0833	10.6613	0.0106
Residuals	6	775.000	129.1667	NA	NA

For the second 1000 revolutions (y_2), all three terms are significant: filler proportion ($p = 0.006$), filler type ($p < 0.001$), and the interaction between them ($p = 0.011$). This indicates that both factors influence wear during this period, and that the effect of filler proportion depends on which filler type is used. The cell means suggest that wear increases markedly with filler proportion for F1, whereas for F2 the wear remains substantially lower and changes relatively little.

Table 9: ANOVA Table for Fabric Wear - Third Period

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Proportion	2	1250.667	625.3333	2.2447	0.1872
Filler	1	24030.750	24030.7500	86.2605	0.0001
Proportion:Filler	2	1064.000	532.0000	1.9097	0.2281
Residuals	6	1671.500	278.5833	NA	NA

For the third 1000 revolutions (y_3), only filler type remains significant ($p < 0.001$); neither filler proportion nor the interaction is significant. By this stage, the dominant factor affecting wear is clearly the type of filler, with F1 producing consistently higher wear than F2 across all proportions.

Table 10: Cell Means for Fabric Wear

Proportion	Filler	y1	y2	y3
P1_25	F1	201.0	190.0	153.0
P2_50	F1	237.0	219.5	186.0
P3_75	F1	267.0	267.5	199.0
P1_25	F2	213.0	116.0	87.5
P2_50	F2	233.5	123.0	94.5
P3_75	F2	234.5	121.0	87.5

Overall, these results suggest that the effects of the factors on fabric wear do change over time. In the early stage, wear is driven mainly by filler proportion; in the middle stage, both proportion and filler type are important, with evidence of interaction; and in the later stage, filler type becomes the primary determinant of wear.

b)

In this context, an interaction between filler type and filler proportion means that the effect of changing the proportion of filler on fabric wear is not the same for the two filler types. In other words, the difference in wear between the filler proportions depends on whether filler F_1 or filler F_2 is being used. This is seen most clearly in the second wear period (y_2), where the interaction term is significant. The cell means show that for F_1 , wear increases substantially as the proportion rises from 25% to 75%, whereas for F_2 , wear remains much lower and changes only slightly across the same proportions. Thus, the impact of filler proportion cannot be described without also specifying the filler type. A significant interaction therefore indicates that the two factors do not act independently on fabric wear but rather that the effect of one factor depends on the level of the other.

c)

If attention is restricted to the final 1000 revolutions (y_3), the results indicate that the most important factor affecting wear is filler type. The ANOVA for y_3 shows a highly significant effect of filler type, while neither filler proportion nor the interaction is statistically significant. This suggests that, in the later stage of the process, the choice between the two fillers has a much greater impact on fabric wear than the proportion used.

From the cell means for y_3 , filler F_2 consistently produces substantially lower wear than filler F_1 at every proportion level. For F_2 , the mean wear values are approximately 87.5, 94.5, and 87.5, compared with 153.0, 186.0, and 199.0 for F_1 . Therefore, if the manager's objective is to minimise fabric wear during the final 1000 revolutions, the clear recommendation is to use filler F_2 . Since the proportion effect is not significant at this stage, there is no strong statistical evidence that one proportion level is better than another, although the 25% and 75% proportions for F_2 give the lowest observed mean wear.